

Netflix Titles Data Report

Analysis of Netflix content structure, trends, and catalog composition

Introduction

This report analyzes the Netflix titles dataset to understand what kinds of content are on the platform, how the catalog has changed over time, and what the data suggests about Netflix's content strategy. Readers should care because the catalog composition reflects broader media trends, including the growth of streaming, audience segmentation, and the balance between movies and series.

Research Question

What does the Netflix catalog look like in terms of content type, release timing, and genre mix, and does the dataset show a meaningful difference between movies and TV shows?

Hypothesis

Null hypothesis: the average release year is the same for movies and TV shows. Alternative hypothesis: the average release year differs between movies and TV shows.

Population of Interest

The population of interest is the full set of Netflix titles represented in this dataset. Because the file contains every observed title in the uploaded catalog, this report treats the data as a census of the available records rather than a random sample of all possible entertainment content.

Sampling Method

No sampling was applied. The uploaded dataset already contains all rows in the file, so every available title was included. If a future project needed to generalize beyond this file, a stratified random sample by content type, release year, and genre would be the strongest approach because it would preserve the structure of the catalog while reducing imbalance across groups.

Bias Identification

This dataset can contain selection bias because it reflects Netflix titles captured at one point in time, not the entire history of every title ever available on the platform. It can also contain missing-data bias, since several fields such as director, cast, country, and date added are incomplete. To reduce bias in interpretation, this report avoids claiming causation, notes missing fields openly, and separates catalog description from population-wide claims.

Dataset / Variables

The dataset contains 8,807 titles and 12 original features: show_id, type, title, director, cast, country, date_added, release_year, rating, duration, listed_in, and description. Additional clean variables were created for analysis: date_added_clean, release_year_clean, type_clean, rating_clean, country_clean, listed_in_clean, movie_duration_min, tv_seasons, primary_genre, and year_added. Type identifies whether a record is a Movie or TV Show; duration stores either minutes or seasons; listed_in provides genre labels; and release_year/date_added track when the title was released and when it was added to Netflix.

Number of samples used

8,807 titles.

Descriptive Statistics

Release year: mean = 2014.18, median = 2017, mode = 2018, minimum = 1925, maximum = 2021.

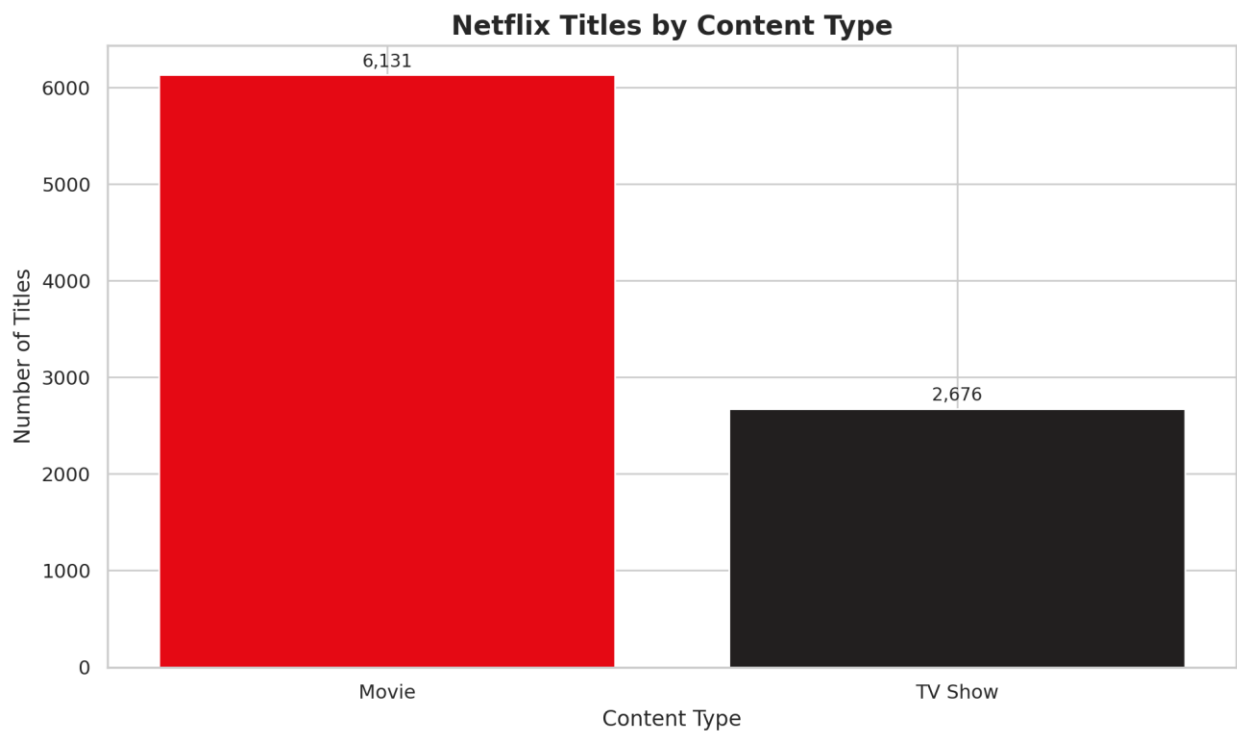
Movie duration: mean = 99.58 minutes, median = 98 minutes, mode = 90 minutes.

TV show length: mean = 1.76 seasons, median = 1 season, mode = 1 season.

The catalog is made up of 6,131 movies (69.6%) and 2,676 TV shows (30.4%).

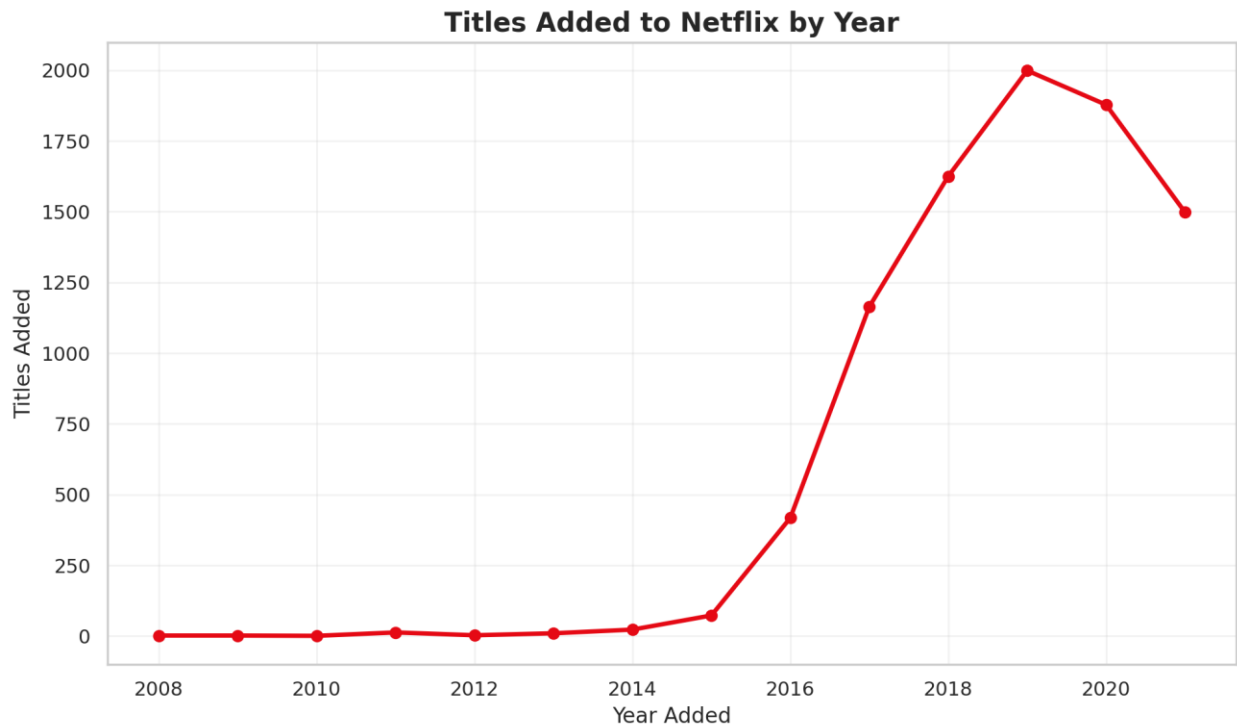
Analysis

Visual 1: Content Type



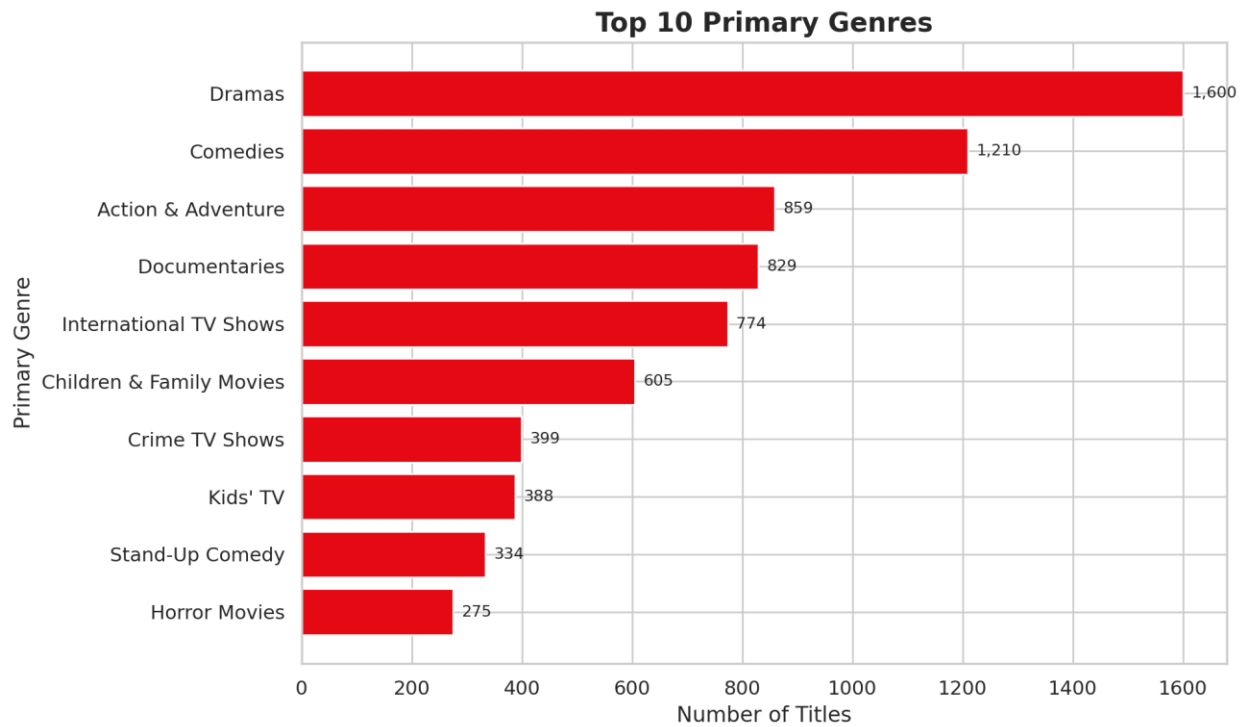
This bar chart shows that movies make up the larger share of the catalog, while TV shows represent a smaller but still substantial portion. This suggests Netflix’s library is movie-heavy overall, although the platform still maintains a large TV content base.

Visual 2: Titles Added Over Time



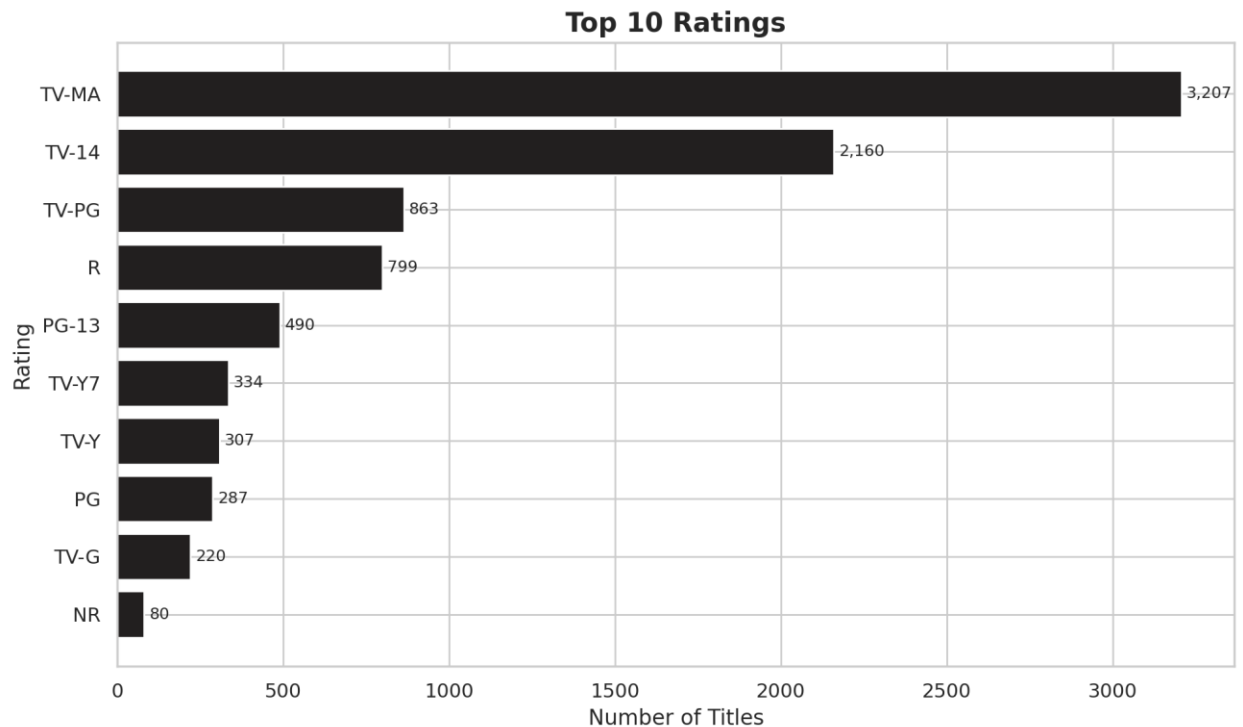
This line chart shows a steep increase in titles added in the late 2010s, with the strongest activity around 2019 and 2020. That pattern indicates Netflix expanded its catalog rapidly during that period and then slowed somewhat in 2021.

Visual 3: Top Genres



This chart shows that dramas, comedies, and action/adventure titles are the most common primary genres. The result suggests Netflix focuses on broad-appeal content categories that are likely to attract wide audiences.

Visual 4: Ratings



This chart shows that TV-MA and TV-14 dominate the rating distribution. That pattern suggests the catalog is skewed toward teen and adult audiences rather than younger children alone.

Hypothesis Testing Steps

Step 1: State the hypotheses. H_0 : the mean release year of movies and TV shows is the same. H_1 : the mean release year differs between movies and TV shows.

Step 2: Choose a significance level. This report uses $\alpha = 0.05$.

Step 3: Run the test. A Welch two-sample t-test produced $t = -20.9763$ and $p = 3.71e-95$. The mean release year for movies was 2013.12, and the mean release year for TV shows was 2016.61.

Step 4: Make a decision. Because the p-value is far below 0.05, the null hypothesis is rejected for this comparison. In this dataset, TV shows are newer on average than movies.

Conclusion

The Netflix catalog in this dataset is dominated by movies, but TV shows are a major part of the platform and appear to be newer on average. The strongest growth in additions occurred around 2019 and 2020, while the genre and rating distributions show a clear focus on drama, comedy, and adult-oriented content. Overall, the dataset suggests a catalog that is broad, recent, and aimed primarily at mature audiences.

Any Potential Issues

Several records are missing values for director, cast, country, and date added, so some analyses cannot use every column. The dataset is also a snapshot of Netflix titles rather than a record of all historical availability, so it should not be treated as a complete causal picture of Netflix strategy. In addition, duration is stored in mixed formats, which required separate handling for movies and TV shows.

Dataset Summary Table

Variable	Description
show_id	Unique title identifier
type	Movie or TV Show
title	Title name
director	Director name, when available
cast	Main cast, when available
country	Country or countries associated with the title
date_added	Date the title was added to Netflix
release_year	Original release year
rating	Audience maturity rating
duration	Runtime in minutes or number of seasons
listed_in	Genre and category labels
description	Short title synopsis